

THARUN RONGALA

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PROFESSIONAL SUMMARY

Aspiring AI Engineer focused on building scalable multi-agent systems and Agentic AI infrastructure, with hands-on learning in LLM orchestration, RAG, and LLMOps deployed on AWS with Docker and Kubernetes. Built and shipped 7+ end-to-end AI/ML projects independently, including two fine-tuning pipelines, while developing strong Python and system design fundamentals through LangChain, LangGraph, LlamaIndex, and the OpenAI/Anthropic APIs.

TECHNICAL SKILLS

Languages and Core: Python, SQL, Git, Data Structures and Algorithms, System Design, Docker, Kubernetes, Microservices, AWS
LLM and Agentic Frameworks: LangChain, LangGraph, LlamaIndex, OpenAI API, Anthropic API, Hugging Face Transformers, PyTorch, TensorFlow

RAG and Retrieval: RAG Pipelines, Vector Databases and Embeddings, ChromaDB, Hybrid Search (BM25 + Dense), Re-ranking, Chunking Strategies

Agentic AI, LLMOps and Evaluation: Multi-Agent Orchestration, Tool Use / Tool Calling, MCP (Model Context Protocol), Structured Outputs, Prompt Engineering, LoRA/QLoRA Fine-Tuning, RAGAS, LangSmith, MLflow, Prometheus, Grafana, FastAPI, Redis, Elasticsearch, Kafka, PySpark

PROJECTS

Agentic Multi-Agent RAG System for Enterprise Knowledge Intelligence

Tech: LangGraph, LangChain, ChromaDB, OpenAI API, FastAPI, Docker, Redis

- Designed and built solo over 4 weeks a LangGraph multi-agent pipeline (planner, retriever, synthesizer) that cut hallucination rates by 62% over direct LLM prompting.
- Hybrid retrieval: ChromaDB dense vectors + BM25 + cross-encoder re-ranker; deployed as a FastAPI/Docker microservice with Redis caching.
- RAGAS: 0.91 context precision, 0.87 answer faithfulness; sub-2-second latency across a 50GB+ document corpus.

Production LLMOps Pipeline with Real-Time Drift Detection and Automated Fine-Tuning

Tech: Apache Kafka, AWS (SageMaker, Lambda, S3), Docker, MLflow, Redis, Prometheus, Grafana, PySpark

- Built a Kafka inference pipeline computing real-time BERTScore and ROUGE drift metrics against baselines to auto-trigger remediation workflows.
- Used PySpark to process and aggregate large-scale inference logs into baseline drift datasets feeding the Kafka pipeline.
- Lambda auto-triggers LoRA fine-tuning on SageMaker, completing a full fine-tuning cycle in about 5 hours on single-GPU resources; MLflow tracks all experiments end-to-end, reducing manual retraining overhead by 80%.
- Prometheus/Grafana dashboards tracked p95 latency and quality scores; 99.7% uptime across 1,000+ concurrent requests over 30 days.

Autonomous AI Research Agent with Multi-Source Knowledge Synthesis

Tech: LangGraph, Anthropic Claude API, Hugging Face Transformers, LlamaIndex, Elasticsearch, Streamlit

- Built an autonomous LangGraph agent (Arxiv, PDF, web tool-calling) integrating MCP servers for external tool access, compressing full literature reviews from hours to under 5 minutes.
- LlamaIndex on Elasticsearch across 10,000+ papers; Anthropic API constitutional loop validates factual consistency before any output.
- Deployed via Streamlit on Hugging Face Spaces; 500+ monthly users, 88% domain QA accuracy, 27 points above zero-shot GPT-4.

CERTIFICATIONS

AI Engineer for Data Scientists Associate | IBM AI Developer Professional Certificate

ACHIEVEMENTS

- Independently built and published 7+ AI/ML/NLP projects on Hugging Face Spaces with Streamlit including 2 LLM fine-tuning projects and 4 additional applied ML/NLP projects gaining hands-on, real-world deployment experience.
- Solved 500+ algorithmic problems on LeetCode and competitive platforms covering data structures, dynamic programming, and graph algorithms.

EDUCATION

National Institute of Technology, Raipur | B.Tech